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Application of Ray Casting Algorithm in Medical Image Enhancement and Edge Detection

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***Abstract*—The Ray Casting algorithm plays an important role in improving medical images and edge detection. Medical imaging typically includes a large amount of image data, such as computed tomography, magnetic resonance imaging, etc. These data need to be effectively and accurately processed and analyzed in order for doctors to make correct diagnoses. The Ray Casting algorithm is a method that allows for effective and accurate processing of medical images. This article mainly used experimental methods and algorithm comparison to analyze the running time and accuracy of medical images using different algorithms. The experimental results showed that the image detection accuracy using the improved Ray Casting algorithm exceeds 95%, which can achieve the enhancement and edge detection of precise medical images, and the effect is better than traditional image processing algorithms. Ray Casting Algorithms can quickly and accurately render medical images, allowing doctors and researchers to better understand and analyze image information. Through the Ray algorithm, the voxel-level enhancement and operation of the medical image are realized to improve the contrast and clarity of the image. To detect and extract edge information in medical images to help assist diagnosis and disease analysis.**

Keywords—ray casting algorithm, medical imaging, edge detection, image enhancement

I. Introduction

Improving medical image and edge recognition has always been an important task in medical image processing. With the continuous development of medical imaging technology, it is necessary to constantly search for better algorithms to process medical imaging data. The Ray Casting algorithm is a beam projection based algorithm suitable for 2D image rendering in 3D scenes. This algorithm is very flexible and scalable, and has been widely used in medical image processing. Medical image enhancement is the process of analyzing and extracting original images through computer image processing technology, which enables them to have high resolution and clearly see the research object visually.

For some edge detection algorithms, due to certain flaws in their principles, they cannot fully guarantee good contrast between the noise and grayscale of the extracted image. There are many theories related to the relationship between Ray Casting algorithm and medical image enhancement and edge detection. For example, some scholars have proposed that in medical image enhancement, some methods need to be used to detect edge features of images [1-2]. Some scholars also believed that the enhancement and edge detection of medical images are complex processes that are interrelated and interdependent [3-4]. Some scholars also suggested that image enhancement can process images to a certain extent, thereby improving image quality [5-6]. In computer image processing, edge detection is usually regarded as an important algorithm. In medical image enhancement, the Ray Casting algorithm can improve the resolution and clarity of images.

This study mainly focused on the application of light projection algorithms in medical image enhancement and edge recognition. The specific content includes the principle and implementation of the light projection algorithm. The ray projection algorithm was used to enhance medical images, including grayscale and tissue compensation. Ray projection algorithms were used to detect edge information in medical images, including edge detection algorithms such as Prewitt and Sobel. This article analyzed and evaluated the effectiveness of the application of light projection algorithms, and analyzed and compared relevant algorithms through experiments. We propose a medical image enhancement method based on Ray Casting algorithm to enhance and improve images by simulating light propagation and interaction. A complete set of medical image edge detection algorithm was developed, which realized the extraction and analysis of edge information in medical image. The medical image enhancement and edge detection system is established, and the Ray Casting algorithm is combined with other technologies to realize the processing and analysis of various image data.

II. Medical Image Enhancement And Edge Detection

A. Medical Image Enhancement

Medical image enhancement detection is a method of improving image quality and obtaining better diagnostic results by enhancing the original medical image, typically including methods such as denoising, contrast enhancement, and image texture enhancement [7-8]. The development of medical image enhancement technology can be divided into several stages: filter based image enhancement technology, and early methods of medical image enhancement use filters to smooth and sharpen images. The selection of folding core size and parameters directly affects the final improvement effect. The enhancement technology based on historical compensation can use the statistical characteristics of image grayscale levels to balance local areas of the image, thereby enhancing the effect [9]. However, this method is prone to excessive improvement. Space domain based extension techniques include region based extension methods, edge based extension methods, etc. These methods can not only improve image contrast, but also improve the peripheral information of the image and help doctors better identify lesion areas. Improved frequency range based technology, where most signals and noise exist in the low frequency range. Therefore, an improved image can be obtained by performing Fourier transform of the image, eliminating low-frequency noise, and then performing inverse transformation [10].

Medical image enhancement and edge recognition are very important technologies in medical image analysis, which can improve the quality and information richness of images and help doctors diagnose diseases more accurately [11]. The implementation of medical image enhancement and edge

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recognition algorithms is usually based on digital image processing technology, mainly including image preprocessing, feature extraction, image enhancement, and boundary recognition steps. In recent years, with the development of deep learning technology, models such as convolutional neural networks have been used to enhance medical images and edge recognition [12]. These models can extract image features and edges more accurately, thereby improving the performance and efficiency of the algorithm. At the same time, it also improves the automation and intelligence of medical image processing. In summary, with the continuous progress of digital image processing technology and deep learning technology, enhancing the role of medical images and edge recognition algorithms in medical image analysis is becoming increasingly important.

B. Edge Detection

In medical image enhancement, edge is a very important aspect, which determines the accuracy of image processing and extraction results. In order to improve the efficiency and anti-interference ability of neural computing, it is necessary to do some work before training to enhance the accuracy and stability of the model. Early medical image enhancement and edge recognition algorithms were mainly based on tissue compensation and filtering techniques, but these techniques had many drawbacks, such as excessive image enhancement or loss of image details. In order to overcome these problems, new algorithms have been developed, such as local contrast enhancement, wavelet transform, Retinex algorithm, etc. These algorithms can improve image quality. At the same time, many edge recognition algorithms have emerged, such as Canny algorithm, Sobel algorithm, Laplace algorithm, etc., which can more accurately recognize the edges of images [13-14].

The implementation of medical image enhancement and edge detection algorithms is mainly achieved through image processing and filtering. However, in practical applications, it is necessary to extract effective information based on the obtained image features. The first step to do is a series of steps such as filtering and grayscale transformation. The image is converted to grayscale values to a certain extent and then compressed using pulse compression technology. The spectrum obtained after preprocessing can be used for edge detection and extraction processes. The required noise signal characteristic parameters and related characteristic indicators are recorded [15-16].

The medical image edge detection method based on Ray Casting algorithm is as follows: preprocess the original medical image, including adjusting grayscale and contrast. The casting algorithm is used to improve preprocessed medical images. By using Sobel, Canny, and other edge detection operators, an initial edge image can be obtained [17]. techniques such as edge refinement and non maximum suppression are used to process the initial image of the edge to obtain the final medical edge. The effectiveness of edge detection is evaluated, including indicators such as accuracy and recall. The medical image edge detection method based on Ray Casting algorithm can effectively improve the analysis and diagnostic performance of medical images, providing important support for clinical applications [18]. The Ray Casting algorithm can serve as the foundation for detecting medical image enhancement. By processing raw medical images, doctors can see lesions more clearly and improve diagnosis.

C. Ray Casting Algorithm

The Ray Casting algorithm is an optical mapping based method that simulates a series of optical phenomena, where light is projected onto a data field and the final rendering result is obtained. Assuming that the 3D data field is filled with small particles, the small particles in the volume data are spheres of equal size and radius. If the number of particles per unit volume is q, the formula for calculating the number of small and medium-sized particles M in a cylindrical thin plate with a cross-section of f is:

$$M = q \times V = q \times F \times \Delta t \tag{1}$$

Among them, V is the volume, denoted by the formula:

$$V = F \times \Delta t \tag{2}$$

The total area T of particles covered by a ray passing through a thin plate is:

$$T = X \times M = X \times q \times F \times \Delta t \tag{3}$$

The Ray Casting algorithm is a three-dimensional image reconstruction method, whose basic idea is to emit a beam of light from the user's viewpoint, track it within the image voxels, calculate its intersection with various locations, and calculate the color and brightness of each pixel based on the intersection position and ray attributes, thereby generating a three-dimensional image. This algorithm was initially used in the field of computer graphics, and also widely used in medical imaging, such as computer tomography, nuclear magnetic resonance imaging, etc. The process of Ray Casting algorithm is shown in Figure 1:

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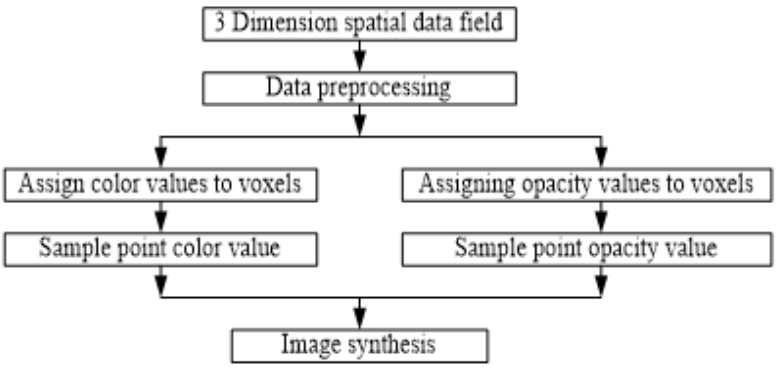

Fig. 1. The Process of Ray Casting Algorithm

The characteristics of light projection algorithms include the ability to generate realistic stereo images, as well as dynamically adjusting lighting, transparency, and other parameters during the rendering process to achieve the best image effect. The characteristics of medical image enhancement testing include the ability to improve diagnostic accuracy, reduce diagnostic omissions and error rates. In the medical field, the combination of ray projection algorithms and improved medical image detection can achieve more accurate, complete, and reliable medical image analysis, and achieve rapid diagnosis and effective treatment of patient diseases. This has important practical value in improving the accuracy and accuracy of medical testing.

Ray Casting is a commonly used image enhancement operator and an enhancement algorithm based on historical compensation. The basic idea is to replace the original pixel gray value with a new pixel gray value obtained through balanced conversion of the original pixel gray level, in order to improve image contrast [19-20]. Specifically, the process of implementing the projection algorithm is as follows: calculating the histogram of the original image using the grayscale values of pixels. The pixel equalization conversion function is calculated based on the histogram of the original image, equalizing each pixel of the original image and replace the grayscale values of the original pixels for analysis of the extended image output. The casting algorithm can effectively improve image contrast, make the details of medical images clearer, and promote precise detection and quantitative analysis.

In terms of medical image enhancement, light projection algorithms can perform multiple light paths and projections on the image, making the image clearer and more vivid. It can calculate the intersection, color, and transparency between each pixel and the target object by performing multiple light projections on the environment of each pixel, making the image more realistic and natural. In the field of medical image edge detection, light projection algorithms can improve the accuracy of patient medical diagnosis by identifying the edges of objects in the image. By projecting each pixel multiple times, this algorithm can accurately calculate the transition, color, brightness, and other information between each pixel and adjacent pixels, making the edges of the object more realistic.

III. Medical Image Enhancement and Edge Detection Experiment

A. Experimental Design

This experiment aims to investigate the application of light projection algorithms in medical image enhancement and edge recognition. This experiment uses a series of human computer tomography images as the experimental dataset, and preprocesses the data, including normalization and denoising, to make it suitable for the application of ray projection algorithms. Ray projection algorithms for data analysis are used to achieve the goals of medical image enhancement and edge detection. The experiment is divided into several parts. Data preprocessing: computer tomography image data preprocessing, including image normalization and noise cancellation processing. Implementation of light projection algorithm: The light projection function of OpenGLO library is used to enhance medical images and edge detection. Improving medical images: Changing different color schemes can improve computer tomography images and make them clearer. Edge detection: The grayscale conversion of medical images makes it more suitable for edge detection. Then, the light projection algorithm is used for edge detection.

The system shall be able to receive different types of medical image data, and shall have the function of pre-processing and format conversion of the data. The module implemented by Ray Casting algorithm should be integrated to enhance and conduct edge detection of medical imaging. An interface for image display and visualization is needed to facilitate doctors and researchers to observe and analyze the processed images. Flexible parameter configuration and adjustment function should be provided to meet the needs of different medical images. High-performance computing and image rendering capabilities are required to achieve real-time performance and interactivity.

B. Experimental Data

The traditional light projection algorithm has the advantage of being able to render high-quality medical images, while allowing interactive operations. It can accurately extract useful information from complex medical records and display structural and surface details. Traditional algorithms use parallel beams of light to intersect with the surface of an object, and calculate the intersection between the beams and the surface through the beam path. Next, the intersection color is used to calculate the pixel color. This algorithm is more accurate and can display more realistic medical images. The disadvantage of traditional light projection algorithms is that they require a large amount of computing resources to process 3D data, which requires high hardware configuration and low rendering efficiency. In large datasets, rendering speed may become very slow and not suitable for real-time applications. The improved algorithm

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can improve the performance of the traditional ray projection algorithm through a variety of methods, such as using graphics processing units for parallel computing, accelerating data access and memory management, and using spatial partitioning technology. Graphics Processing Unit computing can significantly improve processing speed, while spatial partitioning technology can effectively optimize memory management. In addition, machine learning technologies such as deep learning can also be used to improve the accuracy and speed of medical image analysis.

The volume data class provides input data for the Directional Bounding Ray Casting algorithm, including the number of sections scanned by computer tomography, the width and height of each image, and the volume is a pointer to voxels. In the directed boundary radius projection algorithm, a bounding box that works parallel to the coordinate axis is selected. It not only simplifies the beam cutting process, but also facilitates the projection of image layers. In the bounding box class, in addition to displaying the maximum and minimum values of the horizontal, vertical, and spatial axes, there is also an image level bounding box projection function on the imaging plane. The structure of the eight branch tree established in the bounding box mainly has the following parameters. Parent points to the parent node of the node. The children pointer table represents the eight child nodes of a node. The index stores the node's serial number in its sibling nodes. Code stores the encoding information of nodes. Is Leaf a function for storing nodes as leaf nodes and obtaining adjacent nodes.

C. Experimental Environment

The experimental environment for medical image detection needs to include medical image acquisition equipment, corresponding image management and processing software, etc. In order to verify the actual effectiveness and practicality of the Directional Bounding Ray Casting algorithm, the algorithm was implemented on a computer, and multiple sets of data were experimented and compared with traditional algorithms. Experimental data can come from medical imaging datasets such as magnetic resonance imaging and computed tomography, and data processing and preprocessing are required to reduce noise and errors. In the experiment, evaluation indicators need to be set to evaluate the performance of different algorithms, such as error rate, accuracy, recall rate, etc. The experiment also requires analysis methods such as cross validation to evaluate the stability and generalization ability of the algorithm. The specific environment setup is shown in Table 1.

To validate the algorithm proposed in this paper, comparative experiments were conducted using computer tomography image data. The experiment used three sets of body data: skeleton, brain, and head. Ray Casting algorithm (a), ray casting algorithm based on octree coding (b) and Directional Bounding Ray Casting algorithm (c) are used to draw and record the drawing time respectively. The image results generated by different algorithms are shown in Figure 2.

TABLE I. MEDICAL IMAGE DETECTION IN THE EXPERIMENTAL ENVIRONMENT

Environment	Set
Hardware environment	Intel P5 2.40HGz, RADEON 9800 PRO video card, 770M RAM
Software environment	Windows XP, Visual C++5.0 and OPENGLO

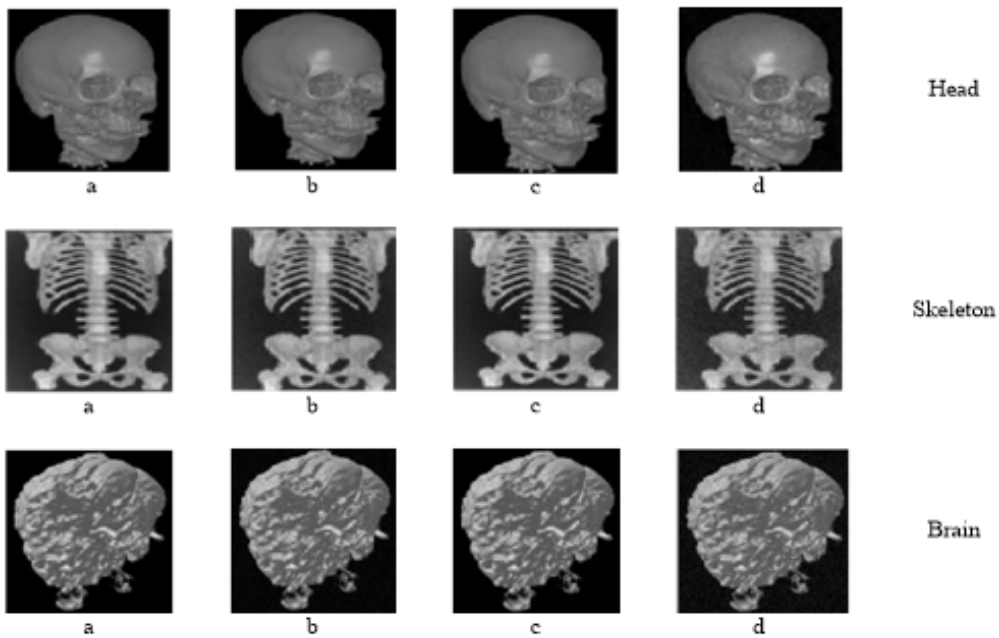

Fig. 2. Comparison of Three Experimental Results

In summary, the traditional Ray Casting algorithm has its unique advantages in medical image rendering, but its performance has certain limitations. Improved algorithms can improve performance through the use of big data technology, parallel computing, deep learning, and other technologies, thus being better applied in the field of medical image analysis and rendering.

IV. EXPERIMENTAL TEST RESULTS

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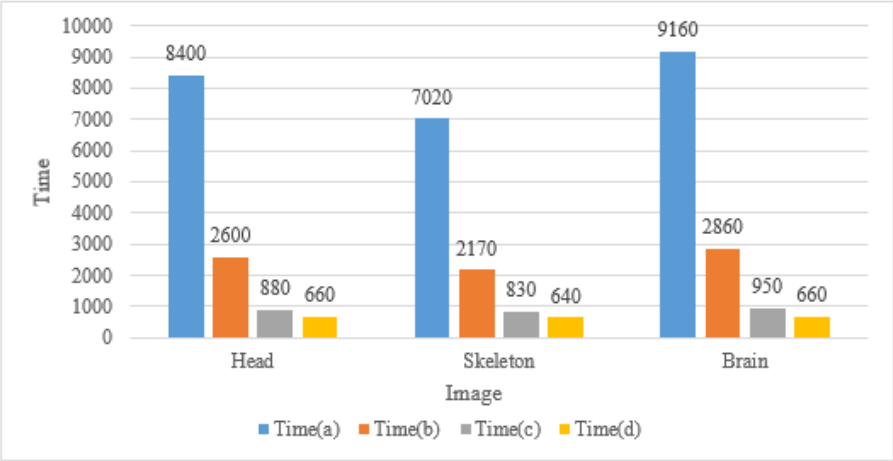

Fig. 3. Time Comparison of Three Experimental Results

As shown in Figure 3, it can be found that in Ray Casting algorithm, the time spent in image a detection of the head is 8400 milliseconds, and the time spent in image b detection by ray casting algorithm based on octree coding is 2600 milliseconds. In the Directional Bounding Ray Casting algorithm, the time for image c detection is 880 milliseconds, and the time for image d detection to run is 660 milliseconds. Ray Casting algorithm in image a detection of bones takes 7020 milliseconds. In general ray casting algorithm based on octree coding, image b detection takes less time, 2170 milliseconds. In the Directional Bounding Ray Casting algorithm, the running time in image c detection is 830 milliseconds, and the running time in image d detection is 640 milliseconds. Ray Casting algorithm spends 9160 milliseconds in image a detection of the brain, and generally the ray casting algorithm based on octree coding spends 2860 milliseconds in image b detection. In the Directional Bounding Ray Casting algorithm, the time spent in image c detection is 950 milliseconds, and the time spent in image d detection is 660 milliseconds.

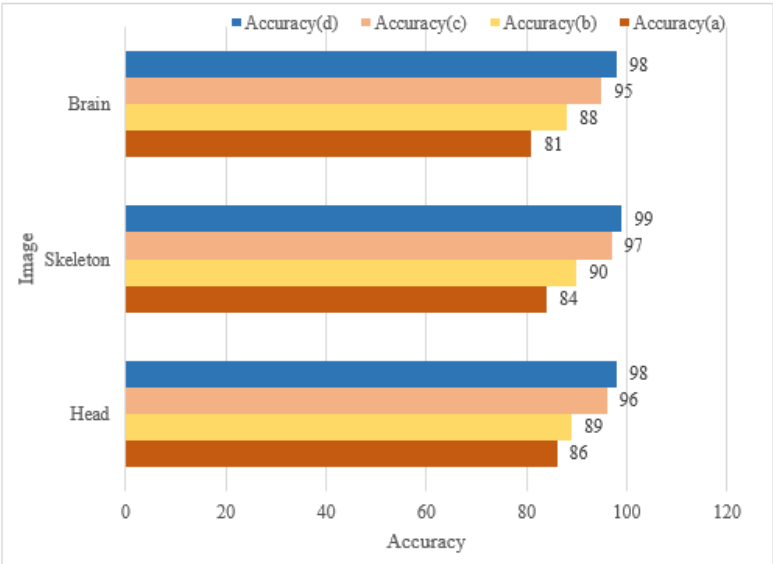

Fig. 4. Accuracy Comparison of Three Experimental Results

Ray Casting algorithm and ray casting algorithm based on octree coding are traditional algorithms, while Directional Bounding Ray Casting algorithm is an improved algorithm. As shown in Figure 4, it can be observed that in traditional algorithms, the accuracy of detecting image a of the head is 86%, and the accuracy of detecting image b is 89%. In the improved algorithm, the accuracy of image c detection is 96%, and the accuracy of image d detection is 98%. The accuracy of image a detection for bones is 84%, and the accuracy of image b detection is 90%. In the improved algorithm, the accuracy of image c detection is 97%, and the accuracy of image d detection is 99%. The accuracy of image a detection for the brain is 81%, and the accuracy of image b detection is 88%. In the improved algorithm, the accuracy of image c detection is 95%, and the accuracy of image d detection is 98%.

By studying the application of ray projection algorithms in medical image enhancement and edge detection, the following conclusions have been drawn. The light projection algorithm can effectively enhance medical images and improve their visual quality. Edge detection using ray projection algorithms can effectively extract edge information from medical images, which is very useful for segmentation and diagnosis of medical images. When using ray projection algorithm to process large format medical image data, it is necessary to

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consider the operational efficiency of the algorithm and the consumption of computing resources. To verify the above conclusion, an experiment was conducted using a set of medical imaging data from computer tomography. The light projection algorithm has been used in experiments for image enhancement and edge detection, and has been compared with traditional image processing algorithms. The experimental results show that the light projection algorithm exhibits excellent performance in image enhancement processing and medical image edge detection. In short, the light projection algorithm is a very valuable medical image processing algorithm that can effectively improve the visual quality and edge recognition effect of medical images.

V. Conclusions

This experience utilized rich knowledge in various fields such as medical image processing, image enhancement, and edge recognition, providing new insights for the research and application of medical images. Due to the light projection algorithm used in this experiment, visualization is clearer and more realistic than other traditional image processing algorithms, meeting the requirements of medical image processing. In addition, different color rendering schemes have different effects in improving medical images and can be selected according to actual needs. Regarding edge detection, the grayscale transformation method used in this experiment effectively extracts edges from images, but in cases of high image processing complexity, problems such as computational errors may arise. Further research and improvement are needed in this regard. Overall, this experiment demonstrates the potential application of ray projection algorithms in medical image processing, thus providing a foundation for future research. For special and complex scenarios, the Ray Casting algorithm may have some blurring or occlusion, and further optimization and improvement of its effectiveness are needed. When processing large-scale medical imaging data, there are high computational resource and time consumption challenges that need to be further improved in algorithm efficiency and optimization. At present, research mainly focuses on the enhancement and edge detection of two-dimensional medical images, and the processing of three-dimensional medical images still needs further research and exploration. The robustness and universality need to be improved, and further research is needed on their adaptability and robustness for different types and sources of medical imaging data.

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